

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA0313	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	Total Marks:	20 marks
CO1 Statement:	Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
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Section / Batch:	A	Date:	05-10-2026

Code of Conduct

I Yoga Madha A/192425058 (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

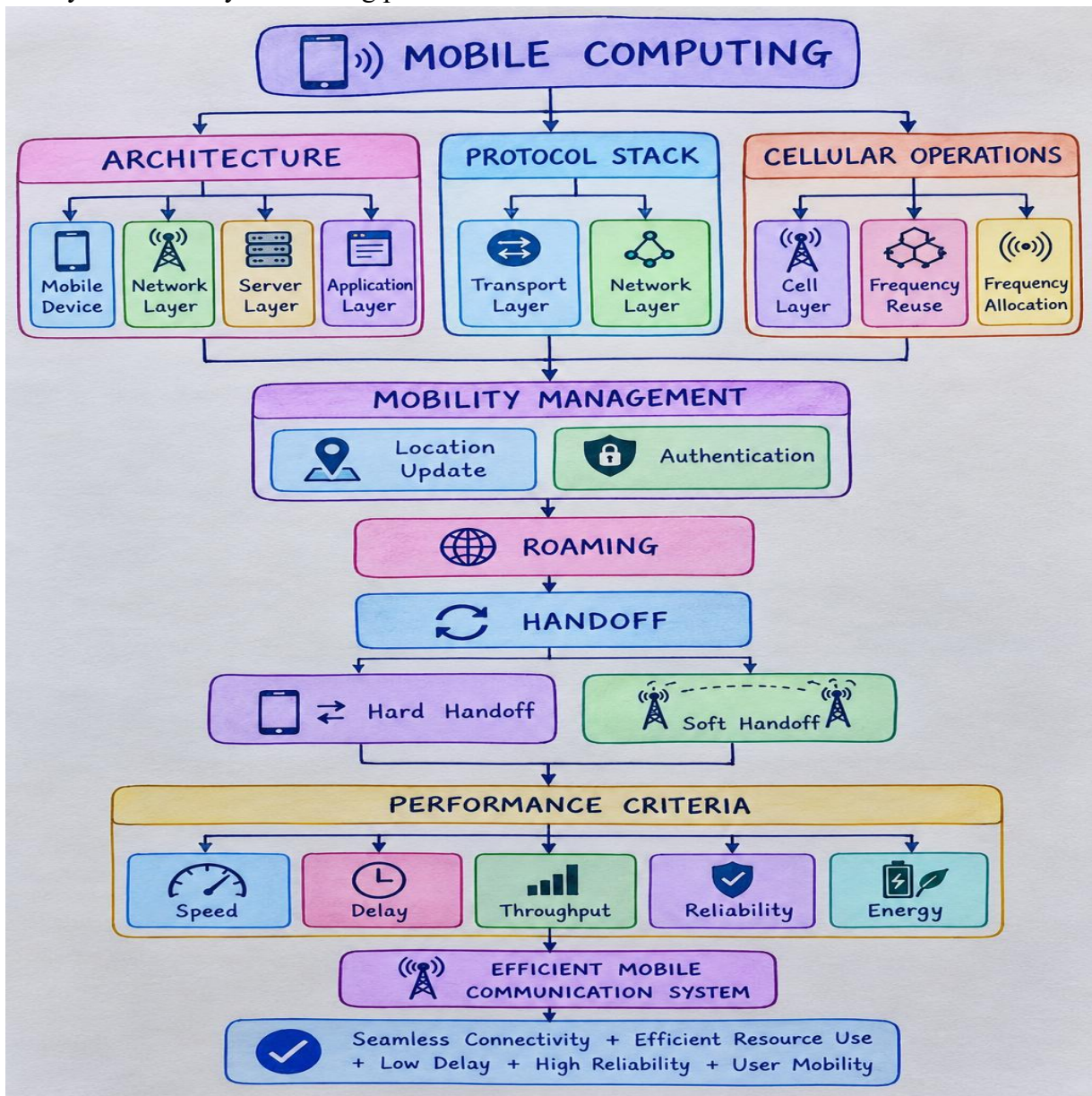
Signature of the Student:



Instructions

- This test contains ONE question.
- Duration: 30 minutes (At home)

- Design a concept map that evaluates the integration of all major concepts in mobile computing. Include architecture, protocol stack, mobility management, roaming, handoff, performance criteria, and cellular operations. Assess how these concepts collectively support an efficient mobile communication system. Justify the hierarchy and linking phrases used.



Explanation of the Hierarchy

- Mobile Computing** is placed at the top because it is the main concept integrating all other concepts.
- Architecture** comes next because it defines the major components such as mobile devices, networks, and servers.
- Protocol stack** supports communication between these components through application, transport, and network protocols.

4. **Mobility management** is connected to the architecture and protocol stack because mobile users can change their location while communicating.
5. **Roaming** is a part of mobility management that allows users to access services outside their home network.
6. **Handoff** follows roaming/mobility because the connection may need to move from one cell or base station to another.
7. **Cellular operations** support mobility through cells, frequency reuse, and resource allocation.
8. **Performance criteria** evaluate whether the complete system is working efficiently.
9. The final goal is an **efficient mobile communication system** providing seamless, reliable, and low-delay communication.

Justification of Linking Phrases

Linking Phrase	Meaning
Mobile Computing → includes → Architecture	Architecture provides the basic structure of the system.
Architecture → uses → Protocol Stack	Protocols enable communication between system components.
Mobile Computing → requires → Mobility Management	Mobile users frequently change locations.
Mobility Management → enables → Roaming	Users can remain connected while moving between networks.
Mobility Management → supports → Handoff	Handoff maintains communication when changing cells.
Cellular Operations → provide → Network Coverage	Cells provide wireless coverage and connectivity.
All concepts → influence → Performance	Architecture, protocols, mobility, and cellular operations affect system performance.
Performance → determines → Efficiency	Speed, delay, throughput, reliability, and energy consumption determine efficiency.

How They Collectively Support an Efficient System

These concepts work together as an integrated system:

Architecture provides the infrastructure → **protocol stack** enables communication → **mobility management** tracks moving users → **roaming** maintains service across networks → **handoff** maintains connectivity between cells → **cellular operations** manage radio resources → **performance criteria** measure efficiency.

Therefore, the integration of these concepts provides **continuous connectivity, efficient resource utilization, high reliability, low latency, and seamless communication for mobile users.**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____